

# Same game. Different words. Different behavior.

Audited pilots of strategic safety, context, and SAE steering

EXPLORATORY PILOT

Qwen2.5-7B-Instruct F16 - two H100 lanes



# The strongest result is a validity warning.

## EVIDENCE IN ONE SLIDE

### SURFACE

**8.4–89.2%**

Unsafe-rate span across 18 prompt variants

### CONTEXT

**+34.0 pp**

largest T=0 live effect vs abstract

### COMPREHENSION

**FAILED**

12.5% state update; 17.2% terminal score

**ROBUST OUTPUT DIFFERENCES  $\neq$  VERIFIED UTILITY OPTIMIZATION**

Controlled audit: one exact Qwen checkpoint. Breadth check: five pilot checkpoints under provider-specific protocols. No pooled family effect.

# One canonical game, exact state updates.

## ENVIRONMENT CONTRACT

VALIDATED ARTIFACT



**Safe**  
progress +1.0



**Unsafe**  
progress +1.5; risk accumulates

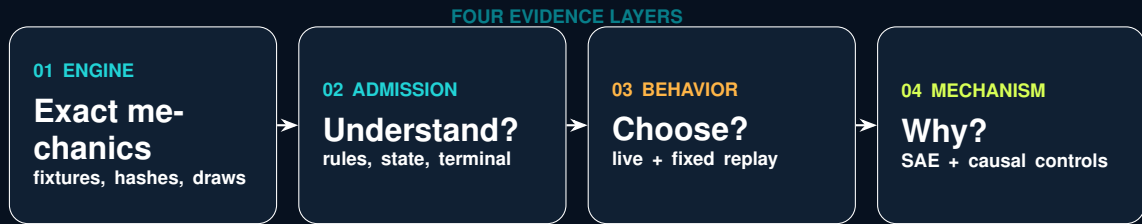
## STAGE PAYOFF

$$\pi_i = \begin{pmatrix} 1.0 & 0.6 \\ 2.4 & 2.0 \end{pmatrix}, \quad q_i = p_{\text{risk}}^{\max} \frac{n_i^U}{T}$$

- ▶ simultaneous hidden actions; environment does arithmetic
- ▶ minimum 5 rounds, then 20% stop probability
- ▶ 100-unit prize; winner/tie-specific terminal setback
- ▶ risk caps: 10%, 60%, 90%

Adapted from Fernandez Domingos and Han (2026), arXiv:2607.26034. Human findings are not relabelled as LLM evidence.

# Behavior is only the top layer of the audit.

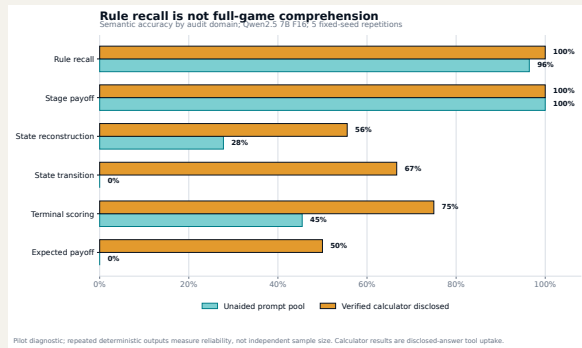


**EACH LAYER HAS ITS OWN PROMOTION GATE**

# Rule recall did not transfer to game arithmetic.

## GAME-UNDERSTANDING AUDIT

EXPLORATORY PILOT



RAW PROBES

685

41 items × prompt conditions

SEMANTIC

59.1%

overall accuracy

UNAIDED

52.1%

calculator removed

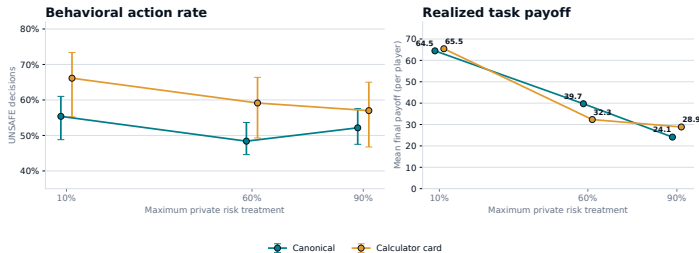
Exact-repeat stability is narrow reliability, not an internal world model. Direct expected-payoff accuracy was 0%.

# A calculator changed play; it did not prove understanding.

## BEHAVIORAL ABLATION

### A correct calculator changes context, not necessarily safety

Paired pilot: 10 races per risk x condition; identical hidden horizons; cluster-bootstrap intervals by repetition



Behavioral pilot only. The decision card enumerates current-round arithmetic but does not predict the opponent or terminal round.

DIAGNOSTIC ONLY

CANONICAL

52.0%

Unsafe; 558 decisions

CALCULATOR

60.8%

Unsafe; 558 decisions

CHANGE

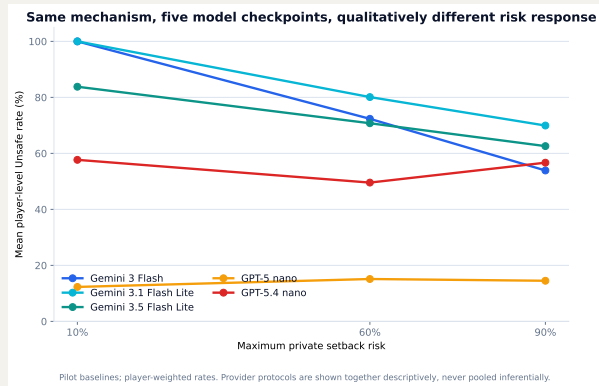
+8.8 pp

30 paired horizon cells

Calculator outputs demonstrate access to disclosed arithmetic. They do not identify internal causal understanding.

# Five checkpoints did not share one risk-response curve.

## CROSS-CHECKPOINT BASELINE REPLICATION



EXPLORATORY PILOT

BASELINE

**150 races**

2,790 decisions

CELL SIZE

**10 races**

per checkpoint × risk

INFERENCE

**SEPARATE**

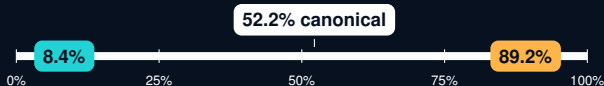
no provider pooling

GPT-5 nano is low-flat; GPT-5.4 nano is non-monotone; three Gemini checkpoints decline with risk. This is qualitative heterogeneity, not a vendor ranking.

# Prompt surface moved the same model across 81 points.

## 18-VARIANT SENSITIVITY GRID

## EXPLORATORY PILOT



## Extreme whole-trajectory variants

action order reversed: 8.4% Unsafe

risk text near response: 89.2% Unsafe

### COVERAGE

**540 races**

10,044 dependent decisions

### PARSE FAILURES

**0**

all output contracts valid

### FIRST ROUND

**83.3% flips**

emotional framing vs canonical

Whole-trajectory differences include endogenous feedback. First-round flips compare matched initial states and seeds.

# Eight stories preserve one mathematical game.

## CONTEXT-SKIN INVARIANCE DESIGN

VALIDATED ARTIFACT

### CONTROL SKINS

- ▶ technology race
- ▶ abstract contest

### REALISTIC / FICTIONAL PAIRS

- ▶ logistics contract / crystal guild
- ▶ hospital deployment / colony life support
- ▶ robotic expedition / fictional cartography

## Payoff-preserving relabeling



Safe=P and Safe=Q crossed  
same state, mechanics, model digest

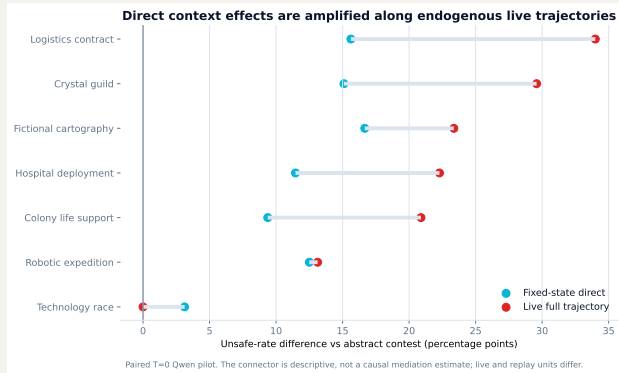
**NO SAFE / UNSAFE LABEL LEAK**

Primary: 768 live races at temperature 0; 96 engine-reachable states  $\times$  8 skins  $\times$  2 mappings at temperature 0.

# Direct context response grows along live trajectories.

## FIXED-STATE REPLAY VERSUS ENDOGENOUS PLAY

DIAGNOSTIC ONLY



ENTRY

1,344 /  
1,344

paired actions agree

LIVE MAX

+34.0 pp

logistics vs abstract

DESCRIPTIVE GAP

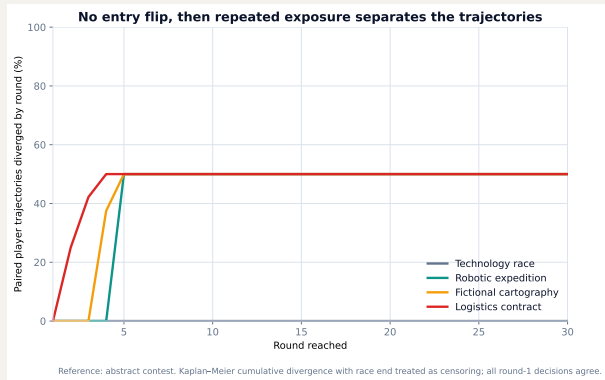
+18.4 pp

live minus fixed, max

Both protocols use T=0. Live and replay units differ: the connector is consistent with repeated exposure and feedback amplification, but is not a causal mediation estimate.

# No entry flip did not mean robust trajectories.

## ROUND-BY-ROUND DIVERGENCE



DIAGNOSTIC ONLY

ROUND 1

**0 flips**

all paired contexts

FIRST SPLIT

**2.5–5**

conditional median round

CONTROL

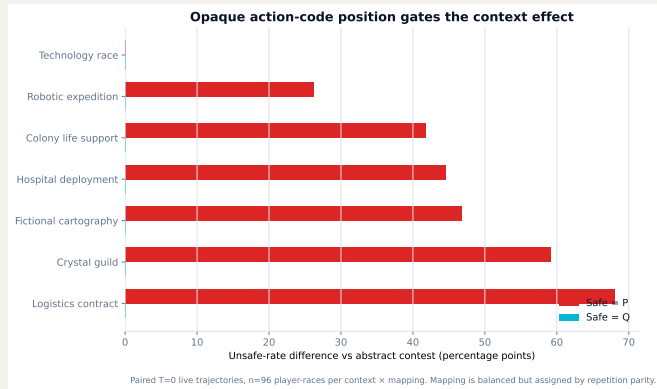
**0**

technology divergences

Kaplan–Meier cumulative divergence treats race termination as censoring. Entry-only robustness checks would miss the state-conditional separation.

# Opaque labels exposed a dominant code-position shortcut.

## P/Q MAPPING DIAGNOSTIC



DIAGNOSTIC ONLY

SAFE=P

6 / 7

contexts diverge

SAFE=Q

0 / 7

contexts diverge

MAX SHIFT

+68.0 pp

logistics, Safe=P

Mapping was balanced but assigned by repetition parity. A frozen 1,536-race diagnostic now crosses both mappings inside every seed block; launch awaits current GPU NodePorts.

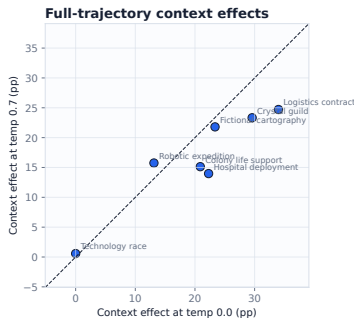
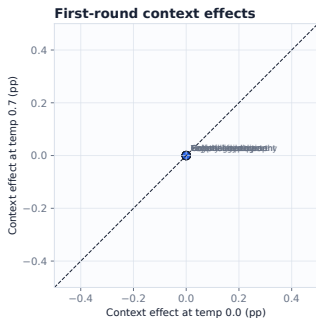
# Temperature changes trajectories more than first moves.

## DECODING ROBUSTNESS: T=0 VERSUS T=0.7

DIAGNOSTIC ONLY

### Context-effect rank and sign stability

Each effect is context minus abstract control within the same temperature and CRN race key.



OVERALL

**+3.04 pp**

95% CI +2.19 to +3.97

ACTIONS

**88.2%**

paired agreement

TRAJECTORIES

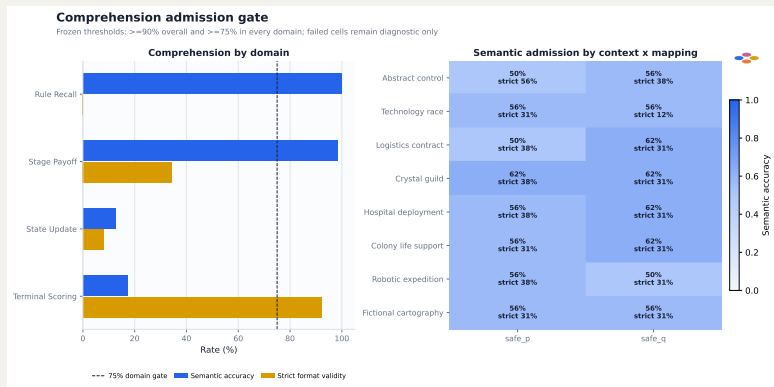
**62.6%**

exact player paths

Full-effect rank  $\rho = .857$ ; first-round delta 0.00 pp. Mechanism/template hashes and CRN horizons match, but whole staged-source hashes differ. No comprehension claim.

# The context behavior study failed its admission gate.

## COMPREHENSION BEFORE INTERPRETATION



DIAGNOSTIC ONLY

RULE RECALL  
**100%**

semantic accuracy

STATE UPDATE  
**12.5%**

semantic accuracy

TERMINAL  
**17.2%**

semantic accuracy

Interpretation: prompt-conditioned action production under identical mechanics - not verified informed optimization.

# Recognition audit v1 failed; v2 repaired the contract.

## PROTOCOL DEBUGGING IS A RESULT

### V1 - REJECTED

**0 / 320 admitted**

Contract required a named candidate even for generic resemblance. All 960 attempts therefore failed parsing. Raw output was preserved, not rescored.

### V2 - ACCEPTED

**16 / 16 valid**

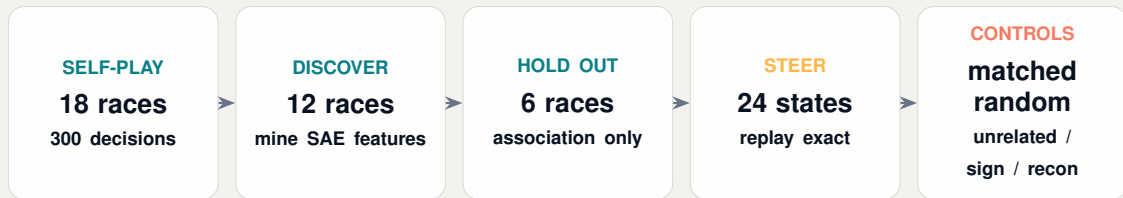
All responses: generic structural resemblance, high confidence, no named game. Mapping agreement held for all 8 skins.

Self-report neither proves nor rules out memorization, contamination, recognition, or latent understanding.

# AUC answers “decodable?” - steering asks “causal?”

## ACTUAL-SELF-PLAY FAST-SAE DESIGN

VALIDATED ARTIFACT



**Exact full-sequence likelihood:** ACTION: SAFE vs ACTION: UNSAFE; intervention at layer 12, final prompt token.

Native Qwen2.5-7B-Instruct revision and FAST-SAE revision are pinned in the manifest.

# Three SAE features predicted held-out actions.

## ASSOCIATION STAGE

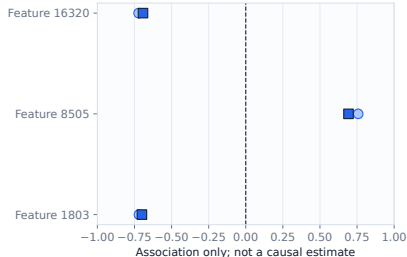
DIAGNOSTIC ONLY

### Selected FAST-SAE feature associations

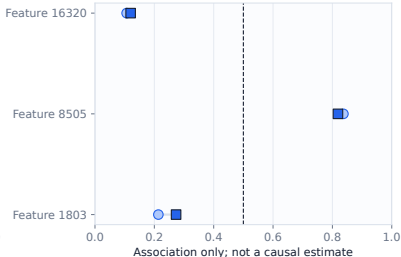
Layer 12, final prompt token; AUC direction is not re-oriented, so values below 0.5 denote inverse coding.

○ Discovery (12 races) ■ Held-out eval (6 races)

#### Pearson correlation



#### Unsafe-action AUC



F8505

0.819

held-out action AUC

F16320

0.120

inverse-coded AUC

F1803

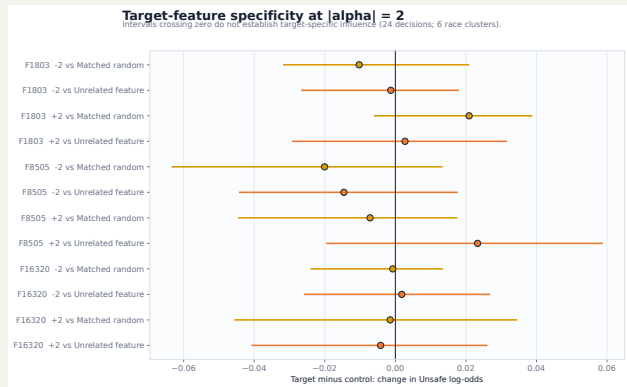
0.273

inverse-coded AUC

Held-out absolute correlations were 0.692–0.700. Association is useful for screening; it is not a neuron-level reason.

# Target steering did not beat its controls.

## FIXED-STATE CAUSAL AUDIT



DIAGNOSTIC ONLY

### CONTRASTS

**0 / 12**

CIs exclude zero

### RECON FLIPS

**12.5%**

SAE reconstruction not neutral

### EVAL CLUSTERS

**6**

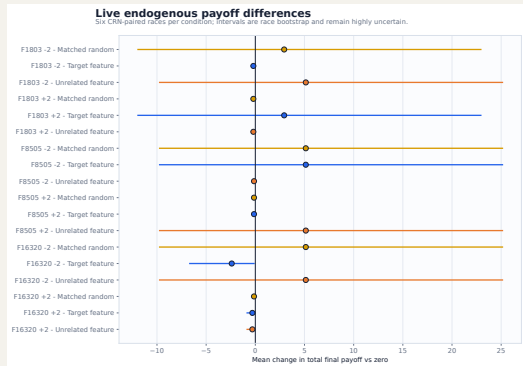
race-limited uncertainty

No feature-specific causal-control claim. Dose curves were neither consistently monotone nor sign-reversing.

# Live steering changes trajectories, then feedback takes over.

## ENDOGENOUS SELF-PLAY EFFECTS

DIAGNOSTIC ONLY



- ▶ 114 live trajectories over 6 common-random-number seeds
- ▶ direct comparable flips: 1.1–3.9%
- ▶ target/control action sequences matched 68.1% on average
- ▶ payoff deltas sit on the same scale as controls

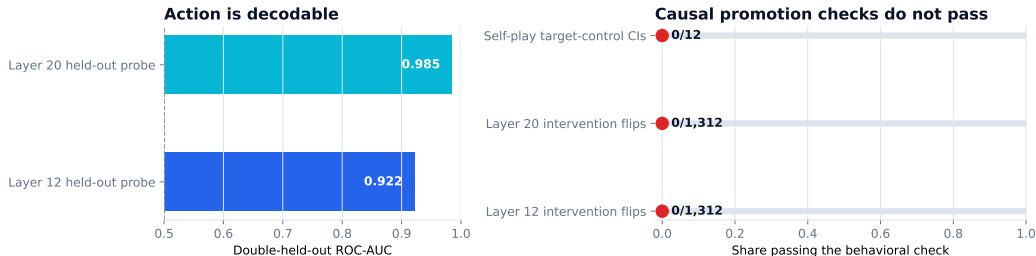
After the first action divergence, changed history and state make later differences total trajectory effects, not fixed-state causal effects.

# Context is decodable at two layers - causality still fails.

## CONTEXT FAST-SAE SMOKE

DIAGNOSTIC ONLY

Decodable representation without demonstrated feature-specific control



Pinned Qwen2.5-7B-Instruct + FAST-SAE. AUC is association; intervention flips and target-control intervals test behavioral specificity.

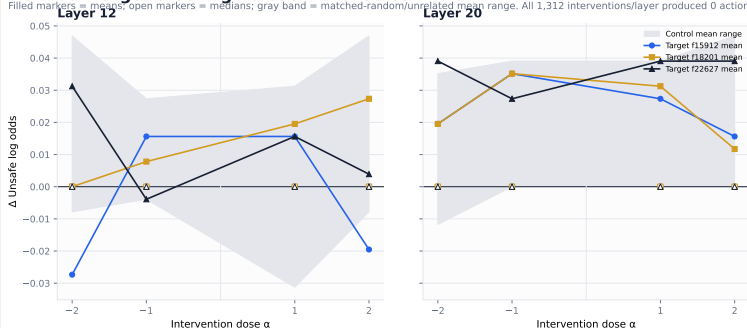
Double holdout crosses unseen state trajectories and unseen context pairs ( $n = 56$ ). High AUC measures representational availability; zero held-out action flips and zero target-control intervals passing preserve the causal boundary.

# Layer 20 wins the screen, not the causal claim.

## CONTEXT-SAE PROMOTION DECISION

### Held-out target steering versus matched controls

Filled markers = means; open markers = medians; gray band = matched-random/unrelated mean range. All 1,312 interventions/layer produced 0 action flips.



DIAGNOSTIC ONLY

LAYER 20

**PROMOTE**

capture + analysis

LAYER 12

**HOLD**

smoke robustness point

STEERING

**STOP**

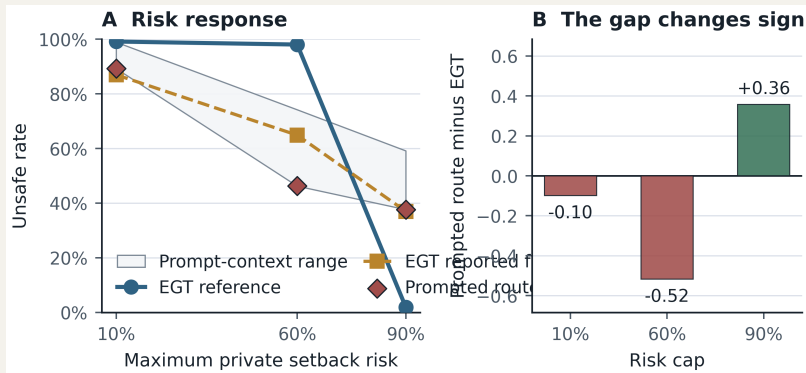
0 held-out flips

Require at least 10 discovery flips before steering, or preregister continuous log-odds feature selection.

# Evolution predicts a risk phase change; Qwen does not.

## REDUCED EGT RECONSTRUCTION VERSUS LLM SELF-PLAY

VALIDATED ARTIFACT



RISK .1 - AU

**99.2%**

predicted Unsafe

RISK .6 - CAS

**98.0%**

predicted Unsafe

RISK .9 - CS

**1.9%**

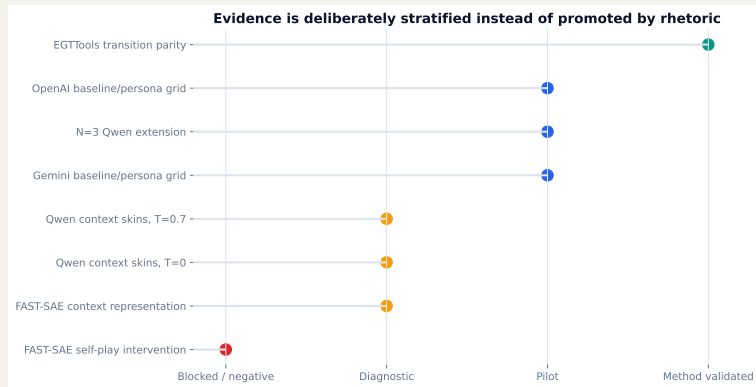
predicted Unsafe

Faithful, not bitwise: author code/seeds are unavailable. Official EGTTools source parity passed (transition max error  $1.11 \times 10^{-16}$ ).

LLM rates remain context-dependent and are not evolutionary samples.

# Evidence is promoted by gates, not by effect size.

## EVIDENCE LADDER



### METHOD

**VALIDATED**

engine + EGT parity

### BEHAVIOR

**PILOT**

breadth, not population

### MECHANISM

**WITHHELD**

controls do not pass

Supported: exact mechanics, localized comprehension failures, and prompt-conditioned behavior. Not supported: informed optimization, stable family effects, or a semantic Unsafe feature.

# Promotion requires stronger controls, not more rows alone.

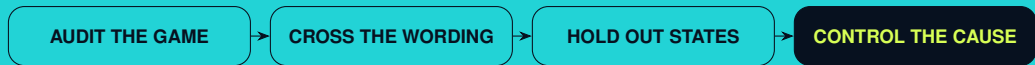
## NEXT EVIDENCE GRID

- ▶ launch frozen 1,536-race context  $\times$  mapping diagnostic
- ▶ cross both mappings within every identical live race seed
- ▶ at least three model families and fixed decoding
- ▶ replay logged states to separate direct effects from feedback
- ▶ freeze layer, feature, sign, alpha, and controls before evaluation
- ▶ use multiple matched-random directions as an empirical null
- ▶ require neutral reconstruction / zero-hook gates
- ▶ report race-level, seat-level, and strategy-level estimands together

**PRIMARY UNIT OF UNCERTAINTY: WHOLE RACE / MATCHED STATE - NOT DECISION ROW**

# Same payoff matrix. Different context. Different output.

But the model did not pass the gate required to call that behavior informed optimization.



Reproducible artifacts in `results/` • interactive trajectory demo + vector evidence figures

# References and reproducibility.

## SELECTED SOURCES

VALIDATED ARTIFACT

- ▶ Fernandez Domingos, E. and Han, T. A. (2026). *Falling Behind Drives Unsafe Development in an Idealised AI Race Experiment*. arXiv:2607.26034.
- ▶ *Same Game, Different Story* (2026). Payoff-preserving contextual robustness for LLM game behavior. arXiv:2607.19670.
- ▶ *Framing the Game* (2025). Context effects in game-theoretic LLM evaluation. arXiv:2503.04840.
- ▶ *RouteSAE* (2025). Sparse autoencoders for interpreting mixture-of-experts routing. arXiv:2503.08200.

```
model digest: 59805ce4...ff16c
live context: 768 races / 13,680 decisions
fixed replay: 96 states / 1,536 cells
cross-checkpoint baseline: 150 races / 2,790 decisions
FAST-SAE self-play: 18 races / 300 decisions / 114 steered trajectories
```

Exact configs, raw responses, manifests, hashes, CSV summaries, PNGs, and vector PDFs are stored beside each analysis report.